

Seminar 6: FIR Filters

Topics: discrete-time systems, moving-average filters, general FIR filters, filter order and length, causality, unit impulses and impulse responses, convolution, unit delays and block-diagram implementations, linearity, time invariance, and cascaded LTI systems.

Key idea. An FIR filter computes each output sample as a finite weighted sum of present and delayed input samples. Its impulse response is the coefficient sequence, and convolution provides the general rule for finding the output to any input.

1. A moving-average filter

Consider the causal three-point moving-average filter $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$.

For the finite input sequence $x[n] = \{2, 4, 6, 4, 2\}$, $0 \leq n \leq 4$,

assume that $x[n]=0$ outside the stated range.

- (a) Compute $y[n]$ for all n for which the output can be nonzero.
- (b) Plot $x[n]$ and $y[n]$ as stem plots and describe the smoothing effect of the moving average.
- (c) Is the filter causal? Explain from the difference equation.
- (d) State the order and length of the filter.
- (e) Write the impulse response $h[n]$ of the filter.

2. From coefficients to an FIR system

An FIR filter has coefficients $b[0] = 3$, $b[1] = -1$, $b[2] = 2$, $b[3] = 1$.

- (a) Write the difference equation relating $y[n]$ to the input samples.
- (b) State the filter order M and filter length L .
- (c) Write $h[n]$ as a finite sequence and also as a sum of shifted unit impulses.
- (d) For the input $x[n]=\{1, 2, 0, -1\}$, with $x[n]=0$ outside $0 \leq n \leq 3$, calculate the complete output sequence.
- (e) Identify the three equivalent descriptions of this FIR filter represented in parts (a)–(c).

3. Unit impulses and decomposition of a sequence

Consider the discrete-time sequence $x[n] = 2\delta[n+1] - \delta[n] + 3\delta[n-2]$.

- (a) Sketch $x[n]$ for $-3 \leq n \leq 4$.
- (b) List the nonzero sample values and their indices.
- (c) Write the shifted sequence $x[n-2]$ as a sum of shifted impulses.
- (d) For a general sequence $x[n]$, write the impulse-decomposition formula that represents $x[n]$ as a sum of scaled, shifted impulses.
- (e) Explain why the impulse decomposition is useful for analysing an LTI system.

4. Impulse response and delayed impulses

A causal FIR system is described by $y[n] = 2x[n] - x[n-1] + \frac{1}{2}x[n-3]$.

- (a) Find the impulse response $h[n]$ by setting $x[n] = \delta[n]$.
- (b) Sketch $h[n]$.
- (c) Determine the output when the input is $\delta[n-4]$.
- (d) Determine the output when the input is $3\delta[n+2]$.
- (e) Explain how the answers to parts (c) and (d) illustrate time invariance and linearity.

5. Numerical convolution

Let $x[n] = \{1, 2, 3, 2\}$, $h[n] = \{2, -1, 1\}$, with both sequences beginning at $n=0$.

- (a) Compute $y[n] = x[n] * h[n]$ using the convolution sum.
- (b) Show the calculation as a sum of scaled and shifted copies of $x[n]$.
- (c) What is the length of the output sequence? Verify it from the input and impulse-response lengths.
- (d) Repeat the convolution in the opposite order, $h[n] * x[n]$, and verify that the same result is obtained.
- (e) State the convolution property illustrated by part (d).

6. Convolution from the impulse response

An LTI system has impulse response $h[n] = \delta[n] + 2\delta[n-1] - \delta[n-2]$.

The input is $x[n] = \delta[n] + \delta[n-2]$.

- (a) Compute the output by applying the system separately to each shifted impulse in $x[n]$.
- (b) Compute the same output directly from the convolution $x[n] * h[n]$.
- (c) Sketch $x[n]$, $h[n]$, and $y[n]$.
- (d) Explain why the two approaches must agree for an LTI system.

7. Direct-form FIR implementation

Consider the third-order FIR filter $y[n] = 2x[n] + x[n-1] - 3x[n-2] + \frac{1}{2}x[n-3]$.

- (a) Draw a direct-form block diagram using unit delays, multipliers, and adders.
- (b) How many unit-delay elements are required?
- (c) How many multipliers and coefficient values are required?
- (d) How many additions are required to combine the four branches?
- (e) Label the signal at the output of each delay element.

8. Causality of FIR systems

For each of the following systems, determine whether it is causal. Give a brief reason in each case.

- (i) $y[n] = x[n] + x[n-1]$,
- (ii) $y[n] = \frac{1}{2}x[n-1] + \frac{1}{2}x[n+1]$,
- (iii) $y[n] = x[n-3]$,
- (iv) $y[n] = x[n] + 2x[n-2] - x[n-5]$.

- (a) For every causal system, identify the largest delay used.
- (b) For the noncausal system, identify the future sample on which the output depends.
- (c) Explain why a centred moving average is generally noncausal when implemented in real time.

(d) Describe one situation in which a noncausal FIR operation can nevertheless be used.

9. Testing linearity and time invariance

Determine whether each system is linear and whether it is time invariant. Justify each answer using the definitions of linearity and time invariance.

- (i) $y[n] = 3x[n] - 2x[n - 1]$,
- (ii) $y[n] = x[n]^2$,
- (iii) $y[n] = n x[n]$,
- (iv) $y[n] = x[n - 2] + 5$,
- (v) $y[n] = x[n] + x[n - 4]$.

(a) Which of these systems are LTI?

(b) For each LTI system, write its impulse response.

(c) Which of the LTI systems are also causal?

10. Using linearity and time invariance without knowing the filter

An unknown LTI system produces the following input-output pair: $x_0[n] \rightarrow y_0[n]$.

Without determining a difference equation or an impulse response, express the outputs for the following inputs in terms of $y_0[n]$.

- (a) $2x_0[n]$.
- (b) $x_0[n-3]$.
- (c) $-x_0[n+2]$.
- (d) $3x_0[n]-2x_0[n-4]$.
- (e) Explain which property of an LTI system is used in each step.

11. The first-difference filter

Consider the FIR filter $y[n] = x[n] - x[n - 1]$.

- (a) Write its impulse response $h[n]$.
- (b) Find the output for the unit-step input $u[n]$.
- (c) Explain why this filter responds strongly at a sudden change in the input but gives zero for a constant sequence away from a transition.
- (d) For $x[n]=\{0,0,2,2,2,-1,-1\}$, beginning at $n=0$, calculate $y[n]$ and identify the locations of the changes.
- (e) State the order and length of the filter.

12. Cascaded FIR systems

Two causal FIR systems are connected in cascade. Their impulse responses are $h_1[n] = \{1,1,1,1\}$, $h_2[n] = \{1,1,1\}$, with both sequences beginning at $n=0$.

- (a) Compute the equivalent impulse response $h_{eq}[n]=h_1[n]*h_2[n]$.
- (b) State the order and length of the equivalent FIR filter.
- (c) Show that interchanging the order of the two systems gives the same equivalent impulse response.
- (d) Explain which property of convolution guarantees the result in part (c).

- (e) If a third FIR system $h_3[n]$ is added to the cascade, explain how associativity allows the equivalent response to be computed in stages.

13. Moving averages, smoothing, and delay

A slowly varying measurement is contaminated by rapid sample-to-sample fluctuations. Two causal moving-average filters are considered:

$$y_3[n] = \frac{1}{3} \sum_{k=0}^2 x[n-k],$$

$$y_7[n] = \frac{1}{7} \sum_{k=0}^6 x[n-k].$$

- (a) State the order and length of each filter.
- (b) Which filter would normally give stronger smoothing? Explain in terms of the number of samples being averaged.
- (c) Estimate the visible delay, in samples, introduced by each causal moving average for a slowly varying waveform.
- (d) Write the corresponding centred three-point average and state whether it is causal.
- (e) Explain the trade-off between stronger smoothing and temporal localization.

14. Equivalent FIR descriptions and implementations

An FIR system has impulse response $h[n] = \delta[n] - 2\delta[n-1] + 2\delta[n-2] - \delta[n-3]$.

- (a) Write the coefficient list $b[0], \dots, b[3]$.
- (b) Write the difference equation.
- (c) Draw a direct-form implementation.
- (d) Explain why any alternative block diagram that produces exactly the same difference equation is an equivalent realization of the filter.
- (e) For an input sequence of length 10, state the maximum length of the complete convolution output.

15. Integrated computational investigation

Use Python with NumPy and Matplotlib to investigate FIR filtering and convolution.

- (a) Create the finite input $x[n] = \{2, 4, 6, 4, 2\}$ and implement causal 3-point and 7-point moving-average filters. Plot the input and outputs.
- (b) For $h[n] = \{3, -1, 2, 1\}$, compute the convolution with $x[n]$ both manually and with `numpy.convolve`, and verify that the results match.
- (c) Generate the unit impulse and verify numerically that filtering it returns the coefficient sequence $h[n]$.
- (d) Implement the first-difference filter and apply it to a piecewise-constant sequence. Plot the detected transitions.
- (e) Convolve two FIR impulse responses in both orders and verify commutativity. Then compare a cascade with the single equivalent FIR filter.
- (f) Summarize how the numerical experiments illustrate impulse response, convolution, causality, smoothing, and the LTI properties.